# Highlights

**Exact-MIP Scheduling with Two-Phase Stability Refinement and Warm-Started Hierarchical Policy Search for Dual-Source Mixed-Flow Rotating-Chamber Cluster Tools**

[TBD: First author], [TBD: Co-author], [TBD: Corresponding author]

- An exact MIP models dual-source mixed-flow rotating-chamber schedules.
- The model adds cleaning, reusable carriers, and finite-horizon decisions.
- A second phase preserves incumbent makespan while reducing instability.
- A shared SPBS warm start supplies feasible incumbents to every solver arm.
- HEM-style cut selection is tested with beam and variable-role structure.

# Exact-MIP Scheduling with Two-Phase Stability Refinement and Warm-Started Hierarchical Policy Search for Dual-Source Mixed-Flow Rotating-Chamber Cluster Tools

[TBD: First author][a], [TBD: Co-author][a], [TBD: Corresponding author][b,*]

[a] *[TBD: Department and University], [TBD: City], [TBD: Postcode], [TBD: Country]*
[b] *[TBD: Department and University], [TBD: City], [TBD: Postcode], [TBD: Country]*

**Abstract**

Dual-source mixed-flow semiconductor cluster tools couple product wafers and reusable process-environment carriers in shared rotating chambers under batching, cleaning, transport, and residency constraints. We first formulate the finite-horizon problem as an exact compact mixed-integer program (MIP). A makespan-first two-phase route uses the remaining budget to preserve the best incumbent makespan, fix most incumbent discrete decisions, and reduce a linear waiting-and-cadence stability measure. A certified primary optimum is claimed only when the first phase is proven optimal; the refinement is not presented as a global lexicographic proof after a time limit. The formulation addresses a different finite-horizon system and adds broader features along selected dimensions than steady-state, single-product, fixed-sequence linear programs for four-finger tools with four-space modules. We then integrate a structure-preserving binary-skeleton (SPBS) primal warm start with a hierarchical cut-selection policy, beam decoding, and SCIP branch-and-cut. The policy inherits the cardinality/ordered-subset decomposition of HEM and adds variable-role descriptors and deterministic role-submodular completion. It only ranks solver-generated valid inequalities and therefore does not alter the scheduling feasible region. A frozen paired protocol is defined to isolate the warm start, hierarchical policy, beam search, and structural features on

*Corresponding author.

*Email address:* `\textcolor {tbdred}{[TBD: corresponding.author@email]}` ([TBD: Corresponding author])

nominal and out-of-distribution instances. [TBD: Insert the validated paired main, component-ablation, and OOD results after fresh checkpoints and all frozen campaigns pass the provenance and solution-validation audits.] The resulting framework couples auditable manufacturing physics with learning-assisted exact optimization without attributing existing HEM components to the present work.

*Keywords:* semiconductor manufacturing, cluster tool scheduling, mixed-integer programming, cutting-plane selection, reinforcement learning, submodular optimization

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## 1. Introduction

Cluster tools integrate several processing modules and robotic handlers around a tightly coupled transport system. They reduce footprint and contamination, but a schedule must coordinate processing, transfers, chambers, load locks, carrier availability, and recipe-dependent temporal restrictions. The difficulty is amplified in a dual-source mixed-flow setting: product wafers and reusable process-environment carriers enter through different sources, meet inside shared chambers, and follow different but interacting routes. Batch and paired processing modes further create discrete choices whose effects propagate through the entire schedule.

Exact MIP is attractive because it can certify schedule quality and encode physical constraints transparently. Its practical limitation is the large branch-and-cut search induced by sequencing and resource-conflict disjunctions. General-purpose solvers generate many candidate cuts at a node, and adding all of them can strengthen the relaxation while also increasing linear-program size, numerical burden, and callback overhead. Cut selection is therefore a sequential decision problem in which local efficacy alone does not capture interactions among cuts.

Recent work has shown that machine learning can support combinatorial optimization without replacing the underlying mathematical model (Bengio et al., 2021). Reinforcement learning has been applied to cut selection (Tang et al., 2020), while look-ahead imitation learning provides another route (Paulus et al., 2022). In particular, HEM already learns both a selected-cut ratio and an ordered subset with a pointer network, and trains the hierarchy with a parallel policy-gradient procedure closely related to A3C. Those elements are treated as the algorithmic baseline here, not as new contributions. Our question is whether a model-derived primal warm start and bounded beam inference, together with formulation roles, improve that baseline when it is embedded in an exact scheduling solver.

This paper studies two linked research questions. First, can an exact finite-horizon model represent dual material sources, mixed $4 \times 1/2 \times 2$ flows, rotating chambers, cleaning, and reusable carriers beyond the steady-state fixed-sequence setting? Second, can SPBS initialization and beam decoding improve the anytime behavior of a HEM-style learned cut selector without changing model validity? We answer them with an exact compact model and an integrated solver, denoted RDMCT-HPS. Variables fixed by manufacturing equalities are continuously represented without relaxing genuine assignment, ordering, mode, or cleaning decisions. The solution stack combines a shared SPBS incumbent, SCIP, a 23-dimensional variable-role hierarchy, and bounded beam decoding.

The paper makes four contributions.

1. We formulate the finite-horizon dual-source mixed-flow cluster-tool problem with shared robots, rotating chambers, two recipe modes, cleaning, carrier residency, and reusable carrier tokens as an exact compact MIP. A status-aware second phase preserves the best incumbent makespan while refining a linear waiting-and-cadence measure under mostly fixed discrete decisions.

2. We construct an SPBS warm start that fixes a physically meaningful assignment/activation/cleaning skeleton and asks SCIP to complete a feasible incumbent. The same incumbent and its construction cost are shared by all algorithm arms in a paired comparison.

3. Relative to an adapted HEM baseline, we retain its hierarchical cardinality and ordered pointer policy, then evaluate bounded beam decoding and 10 formulation-role descriptors with transparent role-submodular completion. The novelty claim concerns this scheduling-specific integration and must be supported by component-wise ablation.

4. We define a frozen, instance-paired evaluation protocol covering model verification, warm-start and search ablations, nominal comparison, factorial conditions, SPBS initialization, sensitivity, and size/recipe distribution shifts.

The current document is an evidence-disciplined draft: numerical cells remain blank until every required paired campaign is complete. This prevents incomplete seeds or successfully solved instances from being reported selectively.

## 2. Related work

### *2.1. Semiconductor cluster-tool scheduling*

Cluster-tool scheduling combines machine scheduling with robotic transport and residency constraints. Exact and analytical methods have been developed for photolithography areas containing cluster tools (Madathil et al., 2018) and for time-constrained tools with purge operations (Yuan et al., 2023). Most directly, Gu et al. (2024) study a four-finger robotic tool with four-space process modules. They analyze steady-state operation under two predetermined cyclic robot sequences—a backward sequence and a hybrid task sequence—and solve one linear program per sequence to minimize cycle time subject to residency constraints. Their paper assumes one product flow and leaves chamber cleaning and simultaneous multiple wafer types to future work. The present model addresses a different handling architecture and adds broader features along selected dimensions: finite-horizon schedules, two material sources, mixed product/carrier flows, flexible assignments and orderings, shared rotating chambers, two recipe modes, explicit cleaning, and reusable carrier tokens. It has not yet been shown to reduce exactly to their four-finger clean/dirty-arm system. We therefore compare modeling coverage and a deliberately constructed special case, not raw runtime between non-equivalent formulations.

### *2.2. Learning within exact MIP*

SCIP provides a modular branch-cut-and-price framework in which learning can act at controlled decision points (Achterberg, 2009; Bestuzheva et al., 2023). Learning-assisted exact optimization is most defensible when the learned component changes search priorities rather than mathematical validity. This principle motivates our callback: it receives only valid cuts generated by SCIP, selects a preferred subset, and leaves omitted cuts available to the solver. The learned model neither creates inequalities nor prunes nodes.

### *2.3. Learned cut selection*

Tang et al. (2020) formulate cut choice as reinforcement learning, while Paulus et al. (2022) use look-ahead information to learn which cuts yield useful bound improvement. Wang et al. (2023) model the number and order of selected cuts hierarchically with 13 generic cut features and a pointer network (Vinyals et al., 2015); their parallel hierarchical policy-gradient training is described as closely related to A3C. We do not claim hierarchical cardinality prediction, ordered subset selection, the pointer network, or parallel sampling as new. We retain that baseline and test three additions: an SPBS primal warm start outside the policy, bounded beam inference at test time, and formulation-role features with deterministic submodular completion. The current implementation's formal configurations use a moving-average REINFORCE baseline, so the training algorithm is called hierarchical policy gradient in the paper. The repository name RDMCT-A3C does not by itself justify a strict asynchronous actor–critic claim.

## 3. Problem description and exact compact formulation

### *3.1. Manufacturing topology*

Product wafers follow the route

$$\mathrm{LP} \to \mathrm{ATR} \to \mathrm{AL} \to \mathrm{ATR} \to \mathrm{LL_U} \to \mathrm{VTR} \to \mathrm{CH} \to \mathrm{VTR} \to \mathrm{LL_L} \to \mathrm{ATR} \to \mathrm{LP},$$

where LP is the load port, ATR and VTR are atmospheric and vacuum transfer robots, AL is the aligner, and $\mathrm{LL_U}$ and $\mathrm{LL_L}$ are the upper and lower load locks. Reusable process-environment carriers (PECs) travel between PEC storage and a shared chamber through the VTR. Chambers CH2 and CH3 rotate and share transfer/load resources.

Two recipe modes are considered. In the $4 \times 1$ mode, four product wafers are processed as a batch using one of two reachable slot groups. In the $2 \times 2$ mode, a PEC and two product wafers form a physically ordered PEC–product–product–PEC chain. Cleaning can be triggered early subject to the cleaning interval; the selected PEC must remain resident for its required chain and is represented by a reusable token. Resource locks prevent simultaneous robot moves and incompatible loading operations.

Table 1 fixes the claim boundary relative to the closest four-space-module study. The rows compare explicitly modeled decisions rather than physical capabilities that are merely present in the equipment.

### *3.2. Event and resource variables*

Let $\mathcal{O}$ be the operations, $\mathcal{R}$ the physical resources, and $\mathcal{A}$ the precedence arcs induced by routes. Each operation $i \in \mathcal{O}$ has start time $s_i$, processing

Table 1: Modeling scope relative to Gu et al. (2024).

| Dimension | Gu et al. (2024) | Present exact model |
|---|---|---|
| Horizon | Steady-state periodic cycle | Finite-horizon event schedule |
| Handling architecture | Four-finger robot with clean/dirty arms | Separate atmospheric/vacuum transfer robots |
| Material flow | One product flow | Product wafers plus reusable PECs |
| Recipes | One flow pattern per case | Mixed $4 \times 1$ and $2 \times 2$ modes |
| Robot decisions | BS or HTS fixed in advance | Assignment and resource order optimized |
| Rotation | Four-space PM rotation delays | Shared rotating chamber/slot-group events |
| Cleaning | Left for future work | Cleaning epochs and early-clean decisions |
| Mathematical program | Two sequence-specific LPs | One exact mixed-integer formulation |
| Objective | Steady-state cycle time | Makespan; conditional incumbent-preserving stability refinement |

duration $p_i$, and completion time $c_i = s_i + p_i$. Binary variables $y_{ijr}$ orient two operations competing for unit-capacity resource $r$. Assignment and mode-selection variables are denoted by $x$, and PEC-token assignments by $u$. The model includes the precedence constraints

$$s_j \geq c_i, \qquad (i,j) \in \mathcal{A}, \tag{1}$$

and resource disjunctions

$$s_j \geq c_i - M_{ijr}(1 - y_{ijr}), \tag{2}$$

$$s_i \geq c_j - M_{jir} y_{ijr}, \qquad \{i,j\} \subseteq \mathcal{O}_r, \tag{3}$$

with event-specific valid constants $M_{ijr}$. Capacity, chamber-rotation, slot-group, cleaning, residency, and token-conservation constraints link $x$, $u$, $y$, and $s$. The complete index-level formulation and parameter construction are provided in [TBD: the online supplement and repository link].

### 3.3. Exact compactness

Some nominally binary membership, slot, and alias variables are already fixed to zero or one by equalities induced by a selected manufacturing mode. We declare these variables continuous in the compact formulation while retaining the fixing equalities. This operation changes neither their values nor the projection of the feasible region; it only removes unnecessary integrality declarations. All genuine assignments, orderings, mode choices, and cleaning choices remain integer. Accordingly, "compact" denotes an equivalent reformulation rather than a relaxation of manufacturing physics.

### 3.4. Makespan-first two-phase refinement

The primary objective is the interval from the first product departure at LP to the last product return. Phase 1 attempts to solve the exact MIP

$$\min_{z \in \mathcal{F}} C_{\max}(z), \tag{4}$$

within its allocated budget. Let $\widehat{z}$ be its best incumbent, $\widehat{C} = C_{\max}(\widehat{z})$, and $\sigma_1$ its recorded solver status. If no incumbent exists, phase 2 is skipped. Otherwise, most discrete assignment/order values $d \in \mathcal{D}_{\text{fix}}$ are fixed to the incumbent, selected double-gripper decisions remain free, and the refinement imposes

$$z_d = \widehat{z}_d \quad (d \in \mathcal{D}_{\text{fix}}), \qquad C_{\max}(z) \leq \widehat{C} + \epsilon. \tag{5}$$

The frozen linear stability objective is

$$\min S_{\text{lin}}(z) = \alpha_w \sum_{i \in \mathcal{W}} w_i + \alpha_m w^{\max} + \alpha_c \sum_{j \in \mathcal{C}} \delta_j + \alpha_e \sum_{k \in \mathcal{E}} e_k, \tag{6}$$

where $w_i$ are modeled waiting/idle quantities, $w^{\max}$ is the maximum individual wait, $\delta_j$ are absolute deviations between successive start cadences, and $e_k$ are soft wait-cap excesses. The weights are fixed before evaluation. The confirmatory configuration reserves 60 s and at most 5000 nodes for this refinement and reports $\sigma_1$ and the phase-2 status separately. If $\sigma_1 = \texttt{optimal}$, $\widehat{C}$ is the certified primary optimum; after a time limit it is only the best known incumbent. Because most discrete decisions are fixed, phase 2 is an incumbent-preserving schedule polish, not a claim of globally optimal lexicographic stability even when the primary optimum is known.

## 4. Warm-started hierarchical cut selection and exact solution

### *4.1. Integrated solution stack and SPBS warm start*

The main solution route is

SPBS incumbent → HEM-style policy/beam → SCIP branch-and-cut → two-phase schedule with

SPBS fixes a physically interpretable binary skeleton comprising batch and chamber assignments, activation indicators, cleaning epochs, slot choices, and selected resource orders. A short feasibility-focused SCIP completion then fills continuous event times and any deliberately unfixed decisions. The resulting solution is accepted only if SCIP reports it feasible and is passed as a primal start to the main MIP. Warm-start generation is preprocessing: its wall-clock cost, success flag, and solution hash are recorded. Within a paired contrast, every compared method receives the identical solution file; if SPBS fails within its budget, the confirmatory pair is marked incomplete and the warm start is regenerated. It is never silently converted into a no-start observation.

This warm start and the learned cut selector address different bottlenecks. SPBS targets time to the first feasible incumbent, whereas the learned policy orders solver-generated cuts to improve the primal–dual trajectory. Their individual and interaction effects are therefore estimated separately.

### 4.2. Decision interface and learning objective

At a separation round, SCIP supplies a legal candidate pool $\mathcal{P} = \{1, \ldots, n\}$. Candidate $i$ represents a valid inequality $a_i^\top z \leq b_i$. A low-cost efficacy prefilter retains at most 128 candidates. The policy selects no more than 16 cuts; omitted candidates are returned in their original legal order.

The state contains candidate features and solver context. An action consists of a selected cardinality and an ordered subset. Training uses the negative SCIP-reported primal–dual integral as reward,

$$R = -\mathrm{PDI}_{\mathrm{SCIP}}, \tag{7}$$

under the frozen SCIP version, time limit, and initial-gap convention. Lower PDI means better anytime performance. The final supplement will state the version-specific bound normalization and no-incumbent convention explicitly; we do not assume numerical identity with the HEM implementation. With baseline $b$, the policy gradient estimator is

$$\widehat{\nabla_\theta J} = \frac{1}{B} \sum_{\ell=1}^{B} (R_\ell - b_\ell) \nabla_\theta \log \pi_\theta(a_\ell \mid s_\ell), \tag{8}$$

following Williams (1992). Independent training samples are evaluated by parallel workers and aggregated before the update. The frozen configuration uses a moving-average reward baseline, not a learned critic. Accordingly, this implementation is reported as parallel hierarchical policy gradient rather than strict asynchronous advantage actor–critic. The repository name RDMCT-A3C is retained for traceability; a direct A3C claim is conditional on running the separate actor–critic implementation and its prespecified comparison.

### 4.3. Generic and formulation-role features

The generic 13-dimensional vector contains objective parallelism, efficacy, support, integral support, normalized violation, and the mean, maximum, minimum, and standard deviation of cut and objective coefficients.

The structural extension partitions variables into five generic type/role families using SCIP variable type and objective coefficient: binaries, general integers, objective-bearing continuous variables, other continuous auxiliaries, and unclassified variables. It does not infer assignment, cleaning, or chamber semantics from variable names. For cut $i$ and role $k$, coefficient-mass share is

$$m_{ik} = \frac{\sum_{j \in \mathcal{V}_k} |a_{ij}|}{\sum_j |a_{ij}| + \epsilon}. \tag{9}$$

The five shares are augmented by bounded-variable ratio, positive-coefficient ratio, discrete–continuous coupling, dominant-role share, and normalized role entropy. These 10 added ratios are invariant to positive row rescaling up to numerical tolerance and rely on formulation types rather than instance-specific variable names. The full 23-dimensional vector is not claimed to be scale invariant because the inherited 13 features include raw coefficient statistics in the frozen, unnormalized configuration.

### *4.4. Hierarchical policy and beam search*

A count policy first predicts the selected proportion. A pointer-style policy then scores an ordered sequence (Vinyals et al., 2015). The network uses a 128-dimensional hidden representation and one glimpse. Training samples stochastically; evaluation uses either greedy decoding or beam width three. End-of-sequence handling, accumulated log probabilities, and backpointers are shared across learned baselines to ensure that a beam-search comparison does not change candidate pools or accounting. HEM already contains the count and ordered pointer policies; only replacing greedy inference with bounded beam decoding is tested as the search extension here. Beam inference and callback time are charged to the common wall-clock limit.

### *4.5. Role-submodular completion*

Let $E$ be the truncated expansion pool formed from the policy ranking followed by a heuristic tail, $A \subseteq E$ the fixed policy anchors, and $S \supseteq A$ the completed set. For normalized quality $q_i$, rank prior $r_i = 1 - \text{rank}(i)/(|\mathcal{R}|-1)$, role mass $m_{ik}$, and $s_{ij} \in [0,1]$ the clipped cosine similarity of the implementation features, define the normalized demand $d_k \propto \sum_{i \in E} m_{ik}$ with $\sum_k d_k = 1$. Completion maximizes

$$F_E(S) = w_q \sum_{i \in S} q_i + w_p \sum_{i \in S} r_i + w_c \sum_k d_k \sqrt{\sum_{i \in S} m_{ik}} + \frac{w_r}{|E|} \sum_{j \in E} \max_{i \in S} s_{ij}, \tag{10}$$

subject to $A \subseteq S \subseteq E$ and $|S| \leq K$. Default normalized weights are $(w_q, w_p, w_c, w_r) = (0.35, 0.20, 0.25, 0.20)$, the completion pool factor is 2.0, and the anchor ratio is 0.25.

With nonnegative modular terms, demand-weighted concave-over-modular role coverage, and facility-location representativeness on fixed $E$, $F_E$ is monotone submodular. Conditional on the fixed anchors and expansion pool, greedy residual completion satisfies

$$F_E(A \cup S_g) - F_E(A) \geq (1 - 1/e) \max_{|B| \leq K - |A|} \{F_E(A \cup B) - F_E(A)\}, \tag{11}$$

where $B \subseteq E \setminus A$ (Nemhauser et al., 1978). The guarantee does not cover anchor choice, expansion-pool truncation, branch-and-cut runtime, or final MIP quality.

### *4.6. Validity and overhead*

RDMCT-HPS never generates a constraint, changes a coefficient, or declares a node infeasible. Every selected inequality has already passed SCIP's legality checks. Unselected cuts remain available according to the callback contract. Thus the learned component can alter search trajectory and runtime but not the scheduling model's feasible set or the validity of an eventual proof. All callback inference, beam search, and postprocessing time is charged to the common wall-clock limit.

### *4.7. Boundary of the optional RL-SAT route*

The repository also contains an optional A3C–beam–CP-SAT route. It searches chamber assignments and solves a CP-SAT timing abstraction for each candidate. Its "optimal" status certifies only that candidate-specific abstraction; it does not prove global optimality of the Petri MIP or of all chamber assignments. RL-SAT is therefore excluded from the main comparison. It may be reported separately only after feature-parity, independent schedule validation, equal end-to-end time accounting, and certificate-scope checks are complete.

## 5. Experimental design

### *5.1. Frozen benchmark suites*

Table 2 summarizes the frozen suites. Nominal instances are used for the principal comparison. The factorial suite varies wafer count, recipe mix, and processing-time scale. Sensitivity conditions perturb one physical factor at a time. The OOD suite increases problem size and changes recipe balance. No test instance is used to tune hyperparameters.

Table 2: Frozen evaluation suites.

| Suite | Design | Conditions/instances |
|---|---|---|
| Nominal main | Frozen nominal manufacturing instances | 20 |
| Factorial DOE | Wafers $\{8, 16, 24, 32\}$, five recipe ratios, processing scale $\{0.9, 1.0, 1.1\}$ | 60 |
| SPBS | Eligible core instances, no primal start vs. the same automatically generated SPBS solution | 48 pairs |
| Sensitivity | Motion $\{0.8, 1.2\}$, processing $\{0.8, 1.2\}$, PEC capacity 10, cleaning interval $\{3, 10\}$ | 8 |
| OOD | Wafers $\{40, 48, 64\}$ and ratios $\{3{:}1, 1{:}1, 1{:}3\}$ | 9 |

Table 3: Methods in the frozen main comparison. All methods use the same formulation, instance, warm-start file, resource limits, solver seed, and global solver settings. Learned selectors additionally share their efficacy prefilter and selection caps; SCIP and ACS retain their native candidate mechanisms.

| Method | Description |
|---|---|
| SCIP | Default SCIP cut selection. |
| ACS | Deterministic adaptive cut-selection baseline tuned only on the frozen validation set. |
| HEM-13D (adapted) | 13 generic features, hierarchical policy, greedy decoding, no variable-role completion. |
| HEM-13D+beam | Adapted HEM with beam width three. |
| 23D feature only | Generic plus role features, without role-submodular reranking. |
| Structure+greedy | 23D policy with role-submodular completion and greedy decoding. |
| RDMCT-HPS | 23D policy, beam width three, policy anchors, and role-submodular completion. |

### 5.2. Compared methods

The adapted HEM-13D row retains the cardinality and ordered-pointer ideas of Wang et al. (2023), but uses the present efficacy top-128 prefilter, a 16-cut cap, training budget, SCIP version, and scheduling protocol; it is not claimed as a bit-for-bit reproduction. HEM-13D+beam changes only test-time decoding. The 23D and structure rows then isolate variable-role ratios and deterministic completion. A separate warm-start experiment must compare no-start and the same SPBS file under SCIP, HEM-13D, and RDMCT-HPS. The warm–policy interaction is a difference in differences, not a single pairwise contrast. This matrix is required before the integrated SPBS+policy+beam stack can be credited with an improvement. [TBD: Implement and audit the learned-policy none/auto Stage before the confirmatory benchmark; the current SPBS Stage covers SCIP only.]

### 5.3. Training and implementation

The frozen policy-gradient configuration uses five independent training seeds, 60 epochs, Adam learning rate $10^{-4}$ for the lower-level policy, batch size eight, maximum gradient norm 2, eight training samples per epoch, and two parallel sample jobs. The high-level selected-percentage policy uses learning rate $5 \times 10^{-4}$ and the reward baseline is an exponential moving average. Candidate and selection caps are 128 and 16. Because the current training files set online evaluation frequency to zero, the prespecified final checkpoint (epoch 60) is used; the manuscript does not claim validation-based checkpoint selection. [TBD: If strict A3C is claimed, rerun training with the audited actor–critic implementation and add the corresponding value loss, update schedule, and ablation.]

Each training solve is limited to 30 s and 1024 MB, with one separation round at the root node. The learning rate decays by a factor of 0.96 every 10 epochs. These settings, the training-instance hashes, and the exact 15 selected `itr_60.pkl` hashes are frozen before evaluation.

Experiments use SCIP through PySCIPOpt. Main, DOE, SPBS, and sensitivity runs have a 600-s time limit; OOD runs have a 1200-s limit and a uniform 4096-MB SCIP memory limit. The final manuscript will report the SCIP, PySCIPOpt, Python, PyTorch, CUDA, GPU, CPU, operating system, thread count, and the stage-specific memory limits after all campaigns are completed consistently. [TBD: Insert the final reproducibility environment table and repository/archive DOI.]

### *5.4. Metrics and statistical analysis*

The primary metric is PDI. Secondary measures are final primal bound, dual bound, relative gap, incumbent availability, phase-specific status and time, node count, timeout rate, memory-failure rate, and optimal-solution rate. [TBD: Before the confirmatory freeze, either add audited telemetry for first feasible/best-incumbent time, root gap, LP iterations, accepted cuts, selected role coverage, redundancy, and callback overhead, or remove the corresponding mechanism analyses.]

Runs are matched by manufacturing instance, solver seed, and, for learned methods, training seed. For inference, solver/training repeats are first aggregated within each manufacturing instance by the prespecified rule; the manufacturing instance is the sampling unit. We then report the paired median difference, a bootstrap confidence interval over instances, the Wilcoxon signed-rank test where applicable, and Holm-adjusted $p$-values. Timeouts and failures remain in the dataset and are summarized explicitly; they are never deleted to improve a method's apparent performance.

A solver status of `timelimit` is not itself an invalid observation. It means that the common wall-clock budget expired before the required proof; an incumbent may still be feasible and operationally useful. On any suite with unsolved instances, equal capped solving times cannot rank methods. Following Wang et al. (2023), the primary comparison therefore uses PDI and also reports incumbent availability, final gap, bounds, and timeout rate. Rows without incumbents are retained. [TBD: Before the confirmatory freeze, state and test the exact SCIP-version initial-gap/no-incumbent convention used by the final submission analysis; do not introduce an undocumented penalty.]

## 6. Results

### *6.1. Archived model sanity checks (non-confirmatory)*

Table 4 records solver-reported outcomes preserved in the local `_results` archive. They demonstrate that representative pure and mixed-flow model variants produced incumbents and, in these four cases, a zero final gap. They were generated before the frozen campaign, lack the current independent solution-validation manifest, and use different instance parameters. They are therefore model illustrations only and are not used in any statistical or algorithmic claim. The inclusion rule is fixed before reading method timings: one parseable zero-gap row for each distinct archived pure or product–carrier configuration with complete $C_{\max}$, node, and time fields. [TBD: Publish the complete archive-status manifest, including excluded, timelimited, errored, and duplicate rows, in the supplement.]

Table 4: Archived single-instance model checks; values are descriptive and not part of the confirmatory experiment.

| Configuration | Product wafers | Status | $C_{\max}$ | Nodes | Time (s) |
|---|---|---|---|---|---|
| $2 \times 2$ token smoke | 4 | optimal | 285.007 | 972 | 1 |
| $4 \times 1$ only | 17 | optimal | 1002.000 | 1 | 3 |
| $2 \times 2$ only | 17 | optimal | 1244.000 | 211276 | 549 |
| $4 \times 1$ archive | 33 | optimal | 984.729 | 69 | 9 |

One additional archived instance contains all four legacy algorithm labels. All methods report the same proven $C_{\max} = 2234$, while wall-clock times are 91 s (SCIP), 89 s (A3C-only), 125 s (heuristic beam), and 127 s (A3C+beam). This single observation verifies neither a speedup nor a beam benefit; its direction in fact makes a complete paired beam ablation essential. Selectively quoting only the 89-s row would be invalid.

### 6.2. Nominal main comparison

Table 5: Nominal main results. Entries will be computed from 20 instances, five solver seeds, and, for learned methods, five training seeds. Lower is better for PDI, gap, and time; higher is better for optimal rate.

| Method | PDI | Final gap (%) | First feasible (s) | Nodes | Optimal (%) |
|---|---|---|---|---|---|
| SCIP | – | – | – | – | – |
| ACS | – | – | – | – | – |
| HEM-13D (adapted) | – | – | – | – | – |
| HEM-13D+beam | – | – | – | – | – |
| 23D feature only | – | – | – | – | – |
| Structure+greedy | – | – | – | – | – |
| RDMCT-HPS | – | – | – | – | – |

[TBD: Report the complete paired nominal comparison, effect sizes, confidence intervals, adjusted tests, and failure counts. Do not fill from the currently partial shards.]

### 6.3. Ablation and mechanism analysis

Table 6: Prespecified mechanism contrasts.

| Contrast | Isolated mechanism | ΔPDI | Holm $p$ |
|---|---|---|---|
| SCIP+SPBS vs. SCIP (no start) | Warm-start incumbent | – | – |
| HEM-13D+SPBS vs. HEM-13D (no start) | Warm effect under HEM | – | – |
| $\Delta_{\text{HEM,warm}} - \Delta_{\text{SCIP,warm}}$ | Warm–policy interaction (DiD) | – | – |
| HEM-13D+beam vs. HEM-13D | Search width | – | – |
| 23D feature only vs. HEM-13D | Variable-role features (both greedy) | – | – |
| Structure+greedy vs. 23D feature only | Submodular completion | – | – |
| RDMCT-HPS vs. Structure+greedy | Beam with fixed structural rule | – | – |

[TBD: Add cut-pool size, selected-cut role coverage, redundancy, callback-overhead, and root-bound analyses to connect solver outcomes to the proposed mechanism.]

### 6.4. Factorial, SPBS, and sensitivity results

Table 7: Robustness summary over designed manufacturing conditions.

| Study | Conditions | Main effect/contrast | ΔPDI | Adjusted $p$ |
|---|---|---|---|---|
| Factorial DOE | 60 | [TBD: Specify prespecified model term] | – | – |
| SPBS | 48 pairs | Automatic vs. none | – | – |
| Sensitivity | 8 | Worst one-factor degradation | – | – |

[TBD: Report factorial interactions only if prespecified and estimable. SPBS changes primal initialization, not the manufacturing feasible region; report its main effect and learned-policy interaction separately from cut-selection effects.]

Table 8: OOD results by wafer count. All compared methods share the 4096-MB memory limit and the 1200-s wall-clock limit. Inc. denotes the percentage of runs with a validated incumbent; timeout, memory-limit, and runner-error rates are reported separately in the accompanying status table.

| Method | 40 wafers | | 48 wafers | | 64 wafers | |
|---|---|---|---|---|---|---|
| | PDI | Inc. (%) | PDI | Inc. (%) | PDI | Inc. (%) |
| SCIP | – | – | – | – | – | – |
| ACS | – | – | – | – | – | – |
| HEM-13D (adapted) | – | – | – | – | – | – |
| HEM-13D+beam | – | – | – | – | – | – |
| 23D feature only | – | – | – | – | – | – |
| Structure+greedy | – | – | – | – | – | – |
| RDMCT-HPS | – | – | – | – | – | – |

### *6.5. Out-of-distribution generalization*

An engineering campaign completed all 729 planned OOD rows on 2 August 2026 under the 4096-MB/1200-s settings and recorded no runner error. Every row ended at the time limit. SCIP and ACS produced incumbents in all 27 respective rows, but each learned method family produced incumbents in only 27 of 135 rows: only the first of five legacy training checkpoints did so. Moreover, all five checkpoint metadata files record `experiment.seed=1`, so the claimed independent training-seed provenance is not auditable. These 729 rows are excluded from Tables 5 and 8; their completion does not repair the checkpoint defect. [TBD: Replace this engineering audit paragraph with validated statistics from the fresh frozen OOD campaign.]

## 7. Industrial implications

The framework separates three responsibilities. Manufacturing engineers define auditable physical constraints; SCIP retains responsibility for valid inequalities and proof; the learned component allocates a limited cut budget. This separation is relevant when an optimizer must be improved without turning feasibility into a black-box prediction. PDI also reflects the operational value of obtaining good schedules early, even when a proof of optimality is not reached within the dispatching horizon.

[TBD: After the complete experiments, translate the observed PDI and time-to-incumbent improvements into operational implications for the tested wafer volumes. Avoid extrapolating throughput or cost beyond the modeled system.]

## 8. Limitations and threats to validity

First, the model and the LPs of Gu et al. (2024) describe non-equivalent equipment decisions; only coverage and a deliberately simplified special-case cross-check are defensible, not a raw runtime claim. Second, the exactness claim belongs to the MIP feasible region and any solver-reported primary proof. The second phase fixes most incumbent discrete decisions and is not a global lexicographic certificate. Third, the role representation depends on a correct mapping from model variables to formulation roles and has not been validated across unrelated MIPs. Fourth, the submodular guarantee applies to residual set completion, not solver runtime. Fifth, the current main training code is parallel REINFORCE, so a strict A3C claim remains outside the evidence unless the actor–critic route is rerun. Sixth, a wall-clock comparison is hardware-sensitive; warm-start construction, inference, and beam time must be accounted for. OOD memory pressure is an outcome rather than a censored inconvenience. Finally, RL-SAT currently certifies a CP-SAT abstraction rather than the full Petri MIP, and deployment would additionally require uncertainty, rescheduling, equipment faults, and fab-level integration.

## 9. Conclusions

This paper presents an exact compact MIP and an integrated warm-started hierarchical policy-search framework for dual-source mixed-flow rotating-chamber scheduling. The model jointly represents finite-horizon product and PEC flows, mixed recipe modes, cleaning, chamber rotation, and shared robotic resources. A second, status-aware phase preserves the primary incumbent makespan while polishing a linear waiting-and-cadence measure; it is not reported as a global lexicographic proof. The solution route combines an SPBS incumbent, the established HEM count/ordered-subset policy, bounded beam decoding, variable-role features, and SCIP. Its callback operates only on solver-generated valid cuts and therefore preserves model validity. A frozen paired protocol has been established to isolate the model, warm start, policy, and beam mechanisms. [TBD: Insert the final evidence-based conclusion only after fresh checkpoints, independent solution validation, and all paired campaigns pass.]

## CRediT authorship contribution statement

[TBD: Insert author-specific CRediT roles.]


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## Declaration of competing interest

[TBD: Insert the authors' approved declaration.]

## Data and code availability

The repository retains the project name RDMCT-A3C; the paper uses the neutral algorithm label RDMCT-HPS because the frozen training route is policy gradient rather than strict actor–critic A3C. The frozen configurations, instance generators, model implementation, training code, and analysis scripts will be archived at [TBD: repository URL and immutable DOI]. Raw campaign outputs and a machine- readable manifest of software/hardware versions will accompany the archive.


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